Experiment Number: 2 MATLAB Simulation of Cellular Coverage, Cell Splitting and Handoff in a Multi-Cell Network

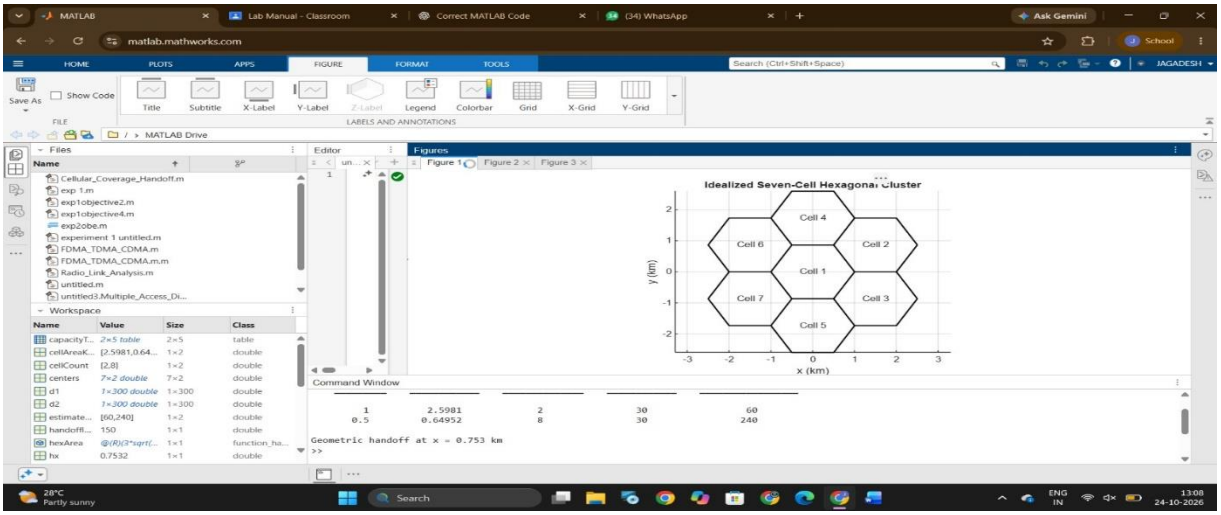
PROGRAM:

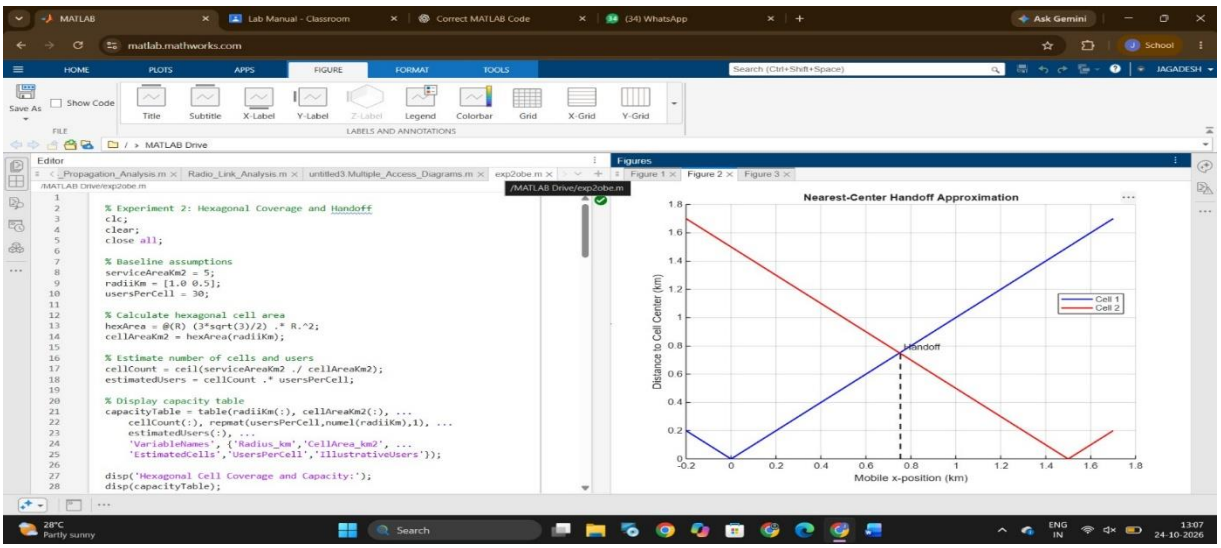
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% Experiment 2: Simplified Hexagonal Coverage and Handoff
clear; clc; close all;
%% 1. Baseline assumptions
serviceAreaKm2 = 5; radiiKm
= [1.0 0.5]; usersPerCell = 30;
% Area of a regular hexagonal cell hexArea =
@(R) (3*sqrt(3)/2) .* R.^2; cellAreaKm2 =
hexArea(radiiKm); cellCount =
ceil(serviceAreaKm2 ./ cellAreaKm2);
estimatedUsers = cellCount .* usersPerCell;
% Display capacity table capacityTable
= table( ...
   radiiKm(:), ...
   cellAreaKm2(:), ...     cellCount(:), ...
repmat(usersPerCell, numel(radiiKm), 1), ...
estimatedUsers(:), ...
   'VariableNames', ...
   {'Radius_km', 'CellArea_km2', 'EstimatedCells', ...
    'UsersPerCell', 'IllustrativeUsers'});

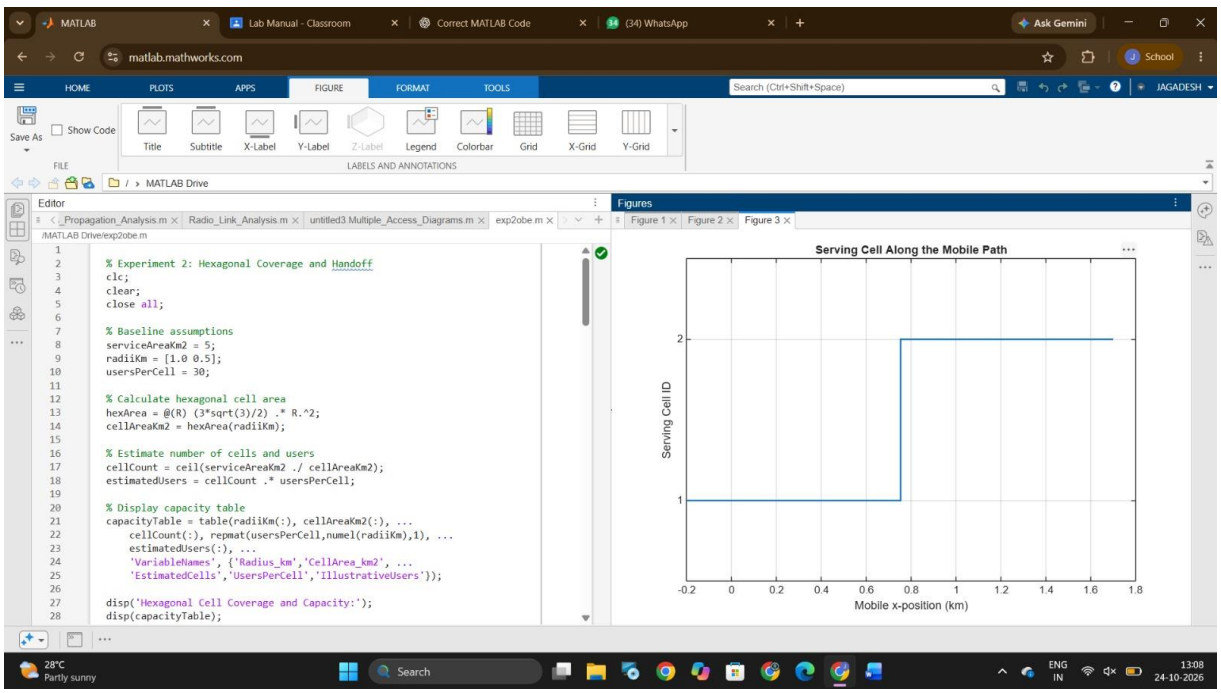
disp('CELLULAR COVERAGE AND CAPACITY');
disp(capacityTable);
%% 2. Seven-cell hexagonal cluster
R = radiiKm(1);
% Flat-top hexagons:
% One central cell and six surrounding cells centers
= R * [ ...
    0,         0;
sqrt(3),     0;
sqrt(3)/2,   1.5;     -
sqrt(3)/2,   1.5;
sqrt(3)/2,  -1.5;     -
sqrt(3)/2,  -1.5;
   -sqrt(3),     0];
theta = (0:6) * pi/3; figure; hold on; axis
equal; grid on; for k = 1:size(centers, 1)
xHex = centers(k,1) + R*cos(theta);
yHex = centers(k,2) + R*sin(theta);
```

```
plot(xHex, yHex, 'k-', 'LineWidth', 1.5);
text(centers(k,1), centers(k,2), ...
       sprintf('Cell %d', k), ...
'HorizontalAlignment', 'center', ...
'VerticalAlignment', 'middle'); end xlabel('x
(km)'); ylabel('y (km)'); title('Idealized Seven-
Cell Hexagonal Cluster'); hold off;
%% 3. Mobile movement between two neighboring cells
cell1 = centers(1,:); cell2 = centers(2,:);
% Mobile moves along the line joining the cell centers
mobileX = linspace(-0.2*R, sqrt(3)*R + 0.2*R, 300);
mobileY = zeros(size(mobileX)); % Distances from
mobile to cell centers d1 = hypot(mobileX - cell1(1),
mobileY - cell1(2)); d2 = hypot(mobileX - cell2(1),
mobileY - cell2(2));
% Assign the nearest cell servingCell =
ones(size(mobileX)); servingCell(d2 < d1) = 2; % Detect
the serving-cell transition handoffIndex =
find(diff(servingCell) ~= 0, 1, 'first'); %% 4. Plot
distances and handoff decision figure; hold on; grid on;
plot(mobileX, d1, 'b-', 'LineWidth', 1.5); plot(mobileX,
d2, 'm-', 'LineWidth', 1.5); if ~isempty(handoffIndex)
xline(mobileX(handoffIndex), '--r', 'Handoff decision');
end xlabel('Mobile x-position (km)'); ylabel('Distance to
cell center (km)');
legend('Cell 1', 'Cell 2', 'Location', 'best');
title('Nearest-Center Handoff Approximation'); hold
off;
%% 5. Plot serving-cell ID along the path
figure; stairs(mobileX, servingCell,
'LineWidth', 1.5); grid on; xlabel('Mobile x-
position (km)'); ylabel('Serving-cell ID');
yticks([1 2]); ylim([0.5 2.5]); title('Serving
Cell Along the Mobile Path');
%% 6. Display handoff position if
~isempty(handoffIndex)     fprintf('Geometric
handoff at x = %.3f km\n', ...
       mobileX(handoffIndex)); else
fprintf('No handoff detected.\n');
end
```

OUTPUT:


Idealized Seven-Cell Hexagonal Cluster
Cell 4
Cell 6
Cell 2
Cell 1
Cell 7
Cell 3
Cell 5
y (km)
x (km)
Geometric handoff at x = 0.753 km


% Experiment 2: Hexagonal Coverage and Handoff
clc;
clear;
close all;
% Baseline assumptions
serviceAreaKm2 = 5;
radiiKm = [1.0 0.5];
usersPerCell = 30;
% Calculate hexagonal cell area
hexArea = @(R) (3*sqrt(3)/2) .* R.^2;
cellAreaKm2 = hexArea(radiiKm);
% Estimate number of cells and users
cellCount = ceil(serviceAreaKm2 ./ cellAreaKm2);
estimatedUsers = cellCount .* usersPerCell;
% Display capacity table
capacityTable = table(radiiKm(:), cellAreaKm2(:), ...
cellCount(:), repmat(usersPerCell,numel(radiiKm),1), ...
estimatedUsers(:), ...
'VariableNames', {'Radius_km','CellArea_km2', ...
'EstimatedCells','UsersPerCell','IllustrativeUsers'});
disp('Hexagonal Cell Coverage and Capacity:');
disp(capacityTable);
Nearest-Center Handoff Approximation
Distance to Cell Center (km)
Mobile x-position (km)
Cell 1
Cell 2
Handoff


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hexArea = @(R) (3*sqrt(3)/2) .* R.^2;
cellAreaKm2 = hexArea(radiiKm);
% Estimate number of cells and users
cellCount = ceil(serviceAreaKm2 ./ cellAreaKm2);
estimatedUsers = cellCount .* usersPerCell;
% Display capacity table
capacityTable = table(radiiKm(:), cellAreaKm2(:), ...
cellCount(:), repmat(usersPerCell,numel(radiiKm),1), ...
estimatedUsers(:), ...
'VariableNames', {'Radius_km','CellArea_km2', ...
'EstimatedCells','UsersPerCell','IllustrativeUsers'});
disp('Hexagonal Cell Coverage and Capacity:');
disp(capacityTable);
Serving Cell Along the Mobile Path
Serving Cell ID
Mobile x-position (km)